=== UNIT 5: LUBRICANTS, REFRACTORIES & PORTLAND CEMENT ===

>> 5.1 Lubrication Mechanisms and Principles

A lubricant is any substance introduced between two moving surfaces in relative contact to reduce friction, minimize wear, and dissipate frictional heat.
* Mechanisms of Lubrication:
1. Thick-Film (Hydrodynamic / Fluid Film) Lubrication: Moving surfaces are completely separated by a continuous, thick fluid lubricant film (~1,000 A). Frictional resistance arises solely from internal fluid shear viscosity: F = eta * A * (dv/dy). Operating in high-speed, light-to-moderate load systems (e.g., sewing machines, hydrodynamic journal bearings).
2. Thin-Film (Boundary) Lubrication: Under high loads or low rotational speeds, the hydrodynamic film squeezes out. Metal asperities come into near-contact, separated only by a chemically or physically adsorbed molecular monolayer (thickness ~1-2 molecules). Governed by polar lubricants (fatty acids, vegetable oils) whose polar headgroups anchor to metal surfaces.
3. Extreme-Pressure (EP) Lubrication: Under severe operating loads and high friction temperatures (>200 deg C), boundary monolayers desorb. EP additives containing active chemical compounds of Sulfur, Chlorine, or Phosphorus react with metal surfaces to form inorganic solid crystalline films (iron sulfide, iron chloride, iron phosphide) with low shear strength, preventing catastrophic seizure and welding.

>> 5.2 Physical and Chemical Properties of Lubricating Oils

* Viscosity and Viscosity Index (VI): Viscosity measures internal resistance to flow. Viscosity Index (Dean & Davis Scale) measures the rate of change of viscosity with temperature:
VI = [(L - U) / (L - H)] * 100, where U = viscosity of test oil at 40 deg C, L = viscosity of zero-VI oil at 40 deg C, and H = viscosity of 100-VI oil at 40 deg C. High VI indicates minimal viscosity change across broad temperature ranges.
* Flash Point and Fire Point: Flash point is the lowest temperature at which an oil gives off sufficient vapor to ignite momentarily when an open flame is applied. Fire point is the temperature at which vapors sustain continuous combustion for at least 5 seconds (typically 5-40 deg C above flash point). Determines operational fire safety.
* Cloud Point and Pour Point: Cloud point is the temperature at which paraffin waxes crystallize, giving a cloudy appearance. Pour point is the lowest temperature at which the oil continues to pour or flow when tilted horizontally. Crucial for sub-zero machinery operation.
* Saponification Value and Acid Number: Saponification value is the mg of KOH required to saponify 1 g of oil (detects animal/vegetable oil blending in mineral oils). Acid number is the mg of KOH required to neutralize free fatty acids in 1 g of oil.

>> 5.3 Refractories and Portland Cement

* Refractories: Heat-resistant inorganic materials capable of withstanding extreme furnace temperatures (>1,580 deg C) without deforming or melting:
- Refractoriness (Pyrometric Cone Equivalent PCE): Tested using Seger cones.
- Refractoriness Under Load (RUL): Temperature at which a refractory specimen collapses under 2.0 kg/cm^2 compressive load.
* Portland Cement: Hydraulic binder consisting of: Tricalcium silicate (C3S, 45-60%, early strength), Dicalcium silicate (C2S, 15-30%, progressive ultimate strength), Tricalcium aluminate (C3A, 6-12%, fast flash setting), and Tetracalcium aluminoferrite (C4AF, 6-8%).
- Setting and Hardening Reactions: Hydration forms tobermorite gel (calcium silicate hydrate C-S-H) and crystalline Ca(OH)2. Gypsum (CaSO4.2H2O, 2-3%) is added during clinker grinding to retard flash setting by forming insoluble calcium sulfoaluminate (ettringite).

[SOLVED PROBLEM 1: Viscosity Index Determination via Dean and Davis Equation]

Problem: A lubricating oil has a kinematic viscosity of 62.0 cSt at 40 deg C and 8.0 cSt at 100 deg C. The reference oils having identical viscosity at 100 deg C (8.0 cSt) exhibit viscosities at 40 deg C of: L = 115.0 cSt (Gulf oil, VI = 0) and H = 55.0 cSt (Pennsylvanian oil, VI = 100). Calculate the Viscosity Index (VI) of the test oil.
Solution Step 1: Identify Parameters:
U (Viscosity of test oil at 40 deg C) = 62.0 cSt.
L (Viscosity of 0-VI standard oil at 40 deg C) = 115.0 cSt.
H (Viscosity of 100-VI standard oil at 40 deg C) = 55.0 cSt.
Solution Step 2: Apply Dean and Davis Formula:
VI = [(L - U) / (L - H)] * 100
L - U = 115.0 - 62.0 = 53.0 cSt.
L - H = 115.0 - 55.0 = 60.0 cSt.
VI = [53.0 / 60.0] * 100 = 0.8833 * 100 = 88.33.
Conclusion: Viscosity Index is 88.33, demonstrating stable thermal viscosity retention suitable for automotive engines.

[SOLVED PROBLEM 2: Saponification Value Determination]

Problem: 2.50 g of a blended lubricating oil was refluxed with 25.0 mL of 0.5 N ethanolic KOH. The unreacted KOH required 12.0 mL of 0.5 N HCl in a back-titration. A blank titration without oil required 24.5 mL of 0.5 N HCl. Calculate the Saponification Value of the oil.

Solution Step 1: Volume of 0.5 N KOH Consumed:
V_consumed = V_blank - V_test = 24.5 mL - 12.0 mL = 12.5 mL of 0.5 N KOH.
Solution Step 2: Saponification Value Formula:
Saponification Value = [V_consumed * Normality of KOH * 56.1] / Mass of Oil (g)
(where 56.1 is molecular weight of KOH in g/mol).
Solution Step 3: Calculation:
Saponification Value = [12.5 * 0.5 * 56.1] / 2.50 = [6.25 * 56.1] / 2.50 = 350.625 / 2.50 = 140.25 mg KOH / g of oil.
Conclusion: Saponification value of 140.25 indicates significant presence of saponifiable triglyceride/vegetable oil blended into the mineral base.

[SOLVED PROBLEM 3: Acid Number Determination of Used Oil]

Problem: 5.00 g of used turbine oil required 3.20 mL of 0.05 N KOH solution for complete neutralization of free fatty acids using phenolphthalein indicator. Calculate the Acid Number and evaluate if oil replacement is required (Replacement threshold: Acid number > 0.5 mg KOH/g).
Solution Step 1: Acid Number Formula:
Acid Number = [V_KOH * N_KOH * 56.1] / Mass of Oil (g)
Solution Step 2: Substitute Values:
Acid Number = [3.20 mL * 0.05 N * 56.1] / 5.00 g = [0.160 * 56.1] / 5.00 = 8.976 / 5.00 = 1.795 mg KOH / g.
Evaluation: The acid value (1.80 mg KOH/g) substantially exceeds the 0.5 mg KOH/g safety limit due to oxidative rancidity and sludge formation. The oil must be replaced immediately to prevent bearing corrosion.

[SOLVED PROBLEM 4: Chemical Composition of Bogue's Compounds in Portland Cement]

Problem: A Portland cement clinker contains the following oxide percentages: CaO = 64.0%, SiO_2 = 21.0%, Al_2O_3 = 6.0%, Fe_2O_3 = 3.0%. Calculate the percentage of Tricalcium Silicate (C_3S) and Dicalcium Silicate (C_2S) using Bogue's empirical equations:
% C_3S = 4.071 * (% CaO) - 7.600 * (% SiO_2) - 6.718 * (% Al_2O_3) - 1.430 * (% Fe_2O_3).
% C_2S = 2.867 * (% SiO_2) - 0.7544 * (% C_3S).
Solution Step 1: Calculate % C_3S:
% C_3S = 4.071 * (64.0) - 7.600 * (21.0) - 6.718 * (6.0) - 1.430 * (3.0)
% C_3S = 260.544 - 159.600 - 40.308 - 4.290 = 260.544 - 204.198 = 56.35%.
Solution Step 2: Calculate % C_2S:
% C_2S = 2.867 * (21.0) - 0.7544 * (56.35) = 60.207 - 42.511 = 17.70%.
Conclusion: Clinker consists of 56.35% C_3S (providing rapid early strength) and 17.70% C_2S (providing progressive long-term hardening).

>> 5.4 University Review & Examination Questions

* Q1: Describe the three mechanisms of lubrication (Thick film, Thin film, and Extreme Pressure) with appropriate operating condition diagrams.
Answer: (1) Hydrodynamic (Thick-Film) Lubrication: Operates under high speeds and low-to-moderate loads. A continuous fluid lubricant film (~1,000 A thick) completely separates opposing metal surfaces. Viscous shear resistance within the lubricant layer replaces solid friction (friction coefficient mu ~ 0.001 - 0.03). (2) Boundary (Thin-Film) Lubrication: Operates under low speeds or high loads where fluid films cannot be maintained. Asperities come into near-contact, separated only by an adsorbed organic monolayer. Long-chain polar molecules (stearic acid, oleic acid) orient perpendicularly, their polar heads bonding electrostatically to the metal surface while hydrocarbon tails project outward (mu ~ 0.05 - 0.15). (3) Extreme-Pressure (EP) Lubrication: Operates under extreme shock loads and high localized temperatures (>200 deg C). Boundary monolayers thermally desorb. Organic compounds of Sulfur, Chlorine, or Phosphorus react chemically with iron to form low-shear-strength solid inorganic layers (FeS, $FeCl_2$) preventing metal-to-metal welding and catastrophic scoring.
* Q2: Explain the setting and hardening of Portland cement, detailing the role of gypsum and hydration reactions of Bogue’s compounds.
Answer: When Portland cement is mixed with water, hydration and hydrolysis reactions occur: (1) Initial Setting: Tricalcium aluminate (C3A) hydrates vigorously within minutes: $C_3A + 6 H_2O \rightarrow C_3A.6H_2O$. This rapid reaction causes premature ’Flash Set’, locking the paste before placement. (2) Role of Gypsum: To prevent flash setting, 2-3% gypsum ($CaSO_4.2H_2O$) is interground with clinker. Gypsum dissolves and reacts with C3A to form an insoluble coating of calcium sulfoaluminate (Ettringite): $C_3A + 3 CaSO_4.2H_2O + 26 H_2O \rightarrow C_3A.3CaSO_4.32H_2O$. This protective needle-like layer retards C3A hydration, extending workability to ~45 minutes. (3) Hardening: Tricalcium silicate (C3S) hydrates within 7 days, providing early compressive strength: $2 C_3S + 6 H_2O \rightarrow C_3S_2.3H_2O$ (Tobermorite C-S-H gel) + $3 Ca(OH)_2$. Dicalcium silicate (C2S) hydrates slowly over months to years, developing progressive ultimate rock-like strength.

>> 5.5 Key Takeaways and Revision Points

* Hydrodynamic lubrication depends on fluid viscosity; Extreme Pressure requires active chemical additives (S, Cl, P).
* Viscosity Index measures resistance to thermal viscosity thinning; high VI is desirable.
* Saponification value indicates animal/vegetable fat blending; acid value measures degradation rancidity.
* Portland cement setting is governed by C3A flash setting retarded by gypsum; C3S and C2S hydration develops structural strength.